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Chemistry faculty perspectives on artificial intelligence and pedagogical change in higher education

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Abstract
This qualitative study investigated chemistry faculty perspectives on the integration of artificial intelligence (AI) in higher education. Three chemistry faculty members from a STEM-focused university participated in the study and provided their perspectives on the use of artificial intelligence. This study was guided by the following research questions: *(1) How do chemistry faculty perceive the role of AI in student learning within their discipline? (2) What are the perceived benefits and challenges of integrating AI technologies into chemistry instruction? (3) How do faculty address ethical considerations associated with AI use in their teaching practices?* Using inductive analysis and open coding, themes were identified directly from participants responses without relying on predetermined categories. The study findings suggest that faculty view AI as a helpful tool for enhancing instructional understanding and increasing student engagement. Participants also discussed concerns regarding academic integrity and other ethical implications related to the use of AI in academic settings. The need for more focused professional development emerged as a positive tool in supporting effective AI integration. The results are presented through a fictionalized data story, highlighting the complexity of faculty experiences with AI.

Keywords Artificial intelligence, Chemistry education, Higher education

1 Introduction

This study examines how chemistry faculty at a STEM-focused university in the southeastern United States interpret and respond to the rapid emergence of artificial intelligence (AI) in higher education. While existing research has explored general faculty attitudes toward AI, comparatively little qualitative work has examined how disciplinary epistemology shapes faculty engagement with AI. Chemistry, as a discipline grounded in experimental verification, error detection, and epistemic accountability, offers a distinct context for examining human–AI collaboration in teaching and learning environments. This study looked at how chemistry faculty were experiencing and responding to the growing presence of generative AI in higher education. Specifically, this study focused on how AI influenced both their teaching practices and the learning experiences of their

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students at a STEM-oriented university. The research was guided by three main questions: *(1) How do chemistry faculty view the role of AI in student learning? (2) What benefits and challenges do they see in bringing AI tools into their teaching? (3) How are they thinking about the ethical issues that come with using AI in the classroom?* To explore these questions, in-depth semi-structured interviews were conducted with three chemistry faculty members. The data was analyzed using an inductive approach, grounded in the principles of grounded theory [1], which allowed themes and insights to emerge directly from the participants' experiences. Grounded theory was a good fit for this work because it helped the researchers understand complex, evolving processes, something especially relevant when exploring faculty responses to such quickly evolving and often controversial technology [2]. Grounded theory offers space for complexity rather than forcing the data into a simple framework. What emerged from the interviews wasn't a clear yes-or-no stance on using AI. Instead, faculty described a more layered and complicated picture, where personal teaching beliefs, departmental expectations, student needs, and comfort levels with technology all played a role in shaping their opinions of AI and its usage. The participant responses were not straightforward, often wavering, situational, and sometimes inconsistent.

The analysis process used open coding, which involved carefully reading through transcripts to identify patterns, themes, and ideas without starting with a fixed set of categories and expectations. While this method is time-consuming and deeply interpretive [1, 3], the researchers took steps to ensure the process was transparent and credible, documenting each step and conducting inter-coder reliability checks to support the trustworthiness of the findings. To make the results more engaging and accessible, they are presented herein in the form of a "data story"—a fictionalized but evidence-based narrative that pulls together real quotes and themes from the interviews. This approach helped to bring faculty voices to life while keeping focus on the underlying patterns in their experiences. This study offers a close look at how the participating chemistry faculty were navigating the opportunities and challenges of AI in the classroom, and what their experiences can teach us about the broader role of technology in STEM education today.

2 A literature overview: AI in higher education

Artificial intelligence (AI) is playing a growing role in higher education, with tools that range from automated content generators to tutoring systems and research aids. Some of the most talked-about developments in recent years are large language models like ChatGPT [4], which can produce text that closely mimics human writing and create images that closely mimic reality based on simple prompts [5]. These capabilities are reshaping how both instructors and students approach learning, teaching, and academic work as a whole [6].

Instructors and researchers alike are currently evaluating how AI tools might be used in the classroom. Several recent studies have explored faculty attitudes, and the findings show a mix of excitement and caution. Faculty members have described AI tools as helpful for increasing efficiency, generating teaching materials, and supporting student engagement [7, 8]. For example, Mukred et al. [8] found that when instructors perceived that AI was easy to use and compatible with their existing teaching styles, they were more likely to adopt it. Concerns have also been raised regarding AI in educational

settings. Particularly around the potential for academic dishonesty, solely relying on AI generated content, and the steep learning curve that comes with mastering new technology [9]. These mixed reactions highlight a rapidly growing concern that's becoming more common in academic spaces. While AI can enhance teaching and improve understanding, it also introduces new ethical and educational challenges that instructors are still navigating.

Another major theme in literature is the importance of faculty preparation. While AI can certainly support learning, its effectiveness often depends on how well instructors understand its limitations and how confidently they can evaluate and differentiate the outputs it produces. Jafari and Keykha [5] and Baskara et al. [7] emphasized that professional development and institutional guidance are crucial if AI is to be used responsibly and meaningfully implemented in educational contexts. A comprehensive review of challenges, opportunities, and implications by Bobula [10] explored both the potential threats that AI poses to higher education, including risks to academic integrity, bias, and misinformation, and its transformative possibilities for pedagogy, institutional strategy, and curriculum.

Recent scholarship increasingly conceptualizes AI integration in education through frameworks of human–AI collaboration, emphasizing negotiated authority, trust, and responsibility between instructors and algorithmic systems. Banihashem et al. [11] and Holmes et al. [12] argue that AI adoption is shaped not only by technical affordances but also by disciplinary norms and epistemological commitments. However, most studies treat STEM disciplines as analytically homogeneous, leaving chemistry's laboratory-centered epistemology underexamined.

Researchers have also explored how students are engaging with AI. Hmoud et al., [13] suggested that tools like ChatGPT can boost student motivation and enjoyment of academic tasks. Others focused on ethical gray areas, such as plagiarism and authorship. For example, Stone [14] found that while many students saw AI tools as useful, they weren't always sure where the ethical lines were when it came to using them for assignments. This uncertainty highlights the need for clear policies and regulations for classrooms regarding the appropriate use of AI tools.

Despite the growing body of work on AI in higher education, there remains noticeable gaps. This is especially true in STEM disciplines like chemistry. Much of the existing research has focused on broader faculty populations or students in general, and a significant number of studies have been conducted outside the United States (U.S.), particularly in Southeast Asia [7, 15]. While these perspectives are valuable, they don't fully capture the unique institutional, cultural, and disciplinary contexts of U.S. based STEM programs. Although Nam and Bai [16] used a comparative media discourse analysis to look at 72 articles from four media outlets, the need to showcase and highlight faculty voice is imperative. In particular, there's very little research on how faculty at STEM focused universities in the Southeastern United States are approaching AI integration. Given the rapid pace of change in both technology and education, it's important to understand how instructors in specific fields—and specific regions—are making sense of these tools and deciding how (or whether) to use them. This study aimed to help fill that gap by focusing on chemistry faculty at a STEM university in the Southeastern United States, adding a local and disciplinary dimension to the broader conversation about AI integration in higher education.

3 Research design & methodology

3.1 Theoretical and methodological framework

This study employed an inductive qualitative design informed by constructivist grounded theory. Rather than generating a formal or substantive theory, grounded theory principles were used as an analytic orientation to produce a discipline-specific conceptual explanation of how chemistry faculty negotiate AI integration within pedagogical, ethical, and epistemic constraints.

Grounded theory was selected as the preferred methodology due to its strength in building theory from lived experience [1]. Grounded theory, as a qualitative research approach, emphasizes theory development through the systematic collection and analysis of data, allowing themes and patterns to emerge organically from the data rather than being dictated by prior theories [17]. Open coding, a key component of grounded theory and inductive analysis, was used to segment and categorize data into discrete parts, enabling the identification of recurring patterns and themes. This iterative process allowed for continuous refinement as new insights emerged [3]. While grounded theory extends beyond open coding to include theoretical sampling and constant comparison [1], open coding served as the foundation for identifying significant concepts and themes directly from the participants' experiences. This approach is particularly well-suited to educational contexts where existing theoretical models may not fully capture the emerging dynamics of new technologies such as AI. As Dey [2] noted, grounded theory allows researchers to uncover layered social processes directly from participant perspectives, which aligned with the goals of this study.

In addition to semi-structured interviews, course syllabi from each participant were systematically reviewed as supplementary documentary data. Syllabi were examined for AI-related policies, assessment language, and academic integrity expectations. These documents were used for triangulation and to contextualize interview claims rather than as independent analytic cases.Data were collected through three semi-structured, in-depth interviews conducted during the Fall 2024 academic term. All interviews were conducted with participant consent and were audio-recorded to maximize transcription accuracy. Interviews lasted between 45 and 60 min and were conducted either in person or via secure video calls, depending on the method preferred by each participant. The interview protocol was designed to encourage open discussion, with questions focusing on experiences with AI, such as their viewpoints on perceived instructional benefits and drawbacks, ethical concerns, and the impact of institutional influences. All participants were assigned pseudonyms to protect confidentiality, and all personal identifying details were removed during the transcription process.

3.2 Setting and participant descriptions

The participants were three full-time chemistry faculty members of similar rank with varying years of teaching experience. The three faculty members were selected based on their willingness to participate in the study, their knowledge of AI tools at varying levels, and their experience in the classroom, making them all well-suited to provide insights into the integration of AI in higher education, specifically in STEM. All three participants were male, worked as full-time tenure-track faculty members, and had different chemistry concentration areas. All three participants were actively involved in research in addition to their teaching responsibilities and had different levels of experience with

AI, from familiarity with AI tools like ChatGPT to limited or exploratory AI usage in the classroom. This diversity allowed for the emergence of varied perspectives on the potential applications and limitations of AI.

3.3 Data analysis

Interview transcripts were analyzed using open coding, following grounded theory principles. The coding process was iterative and inductive, allowing categories and themes to emerge from the data rather than being imposed in advance. Following initial coding, themes were refined through constant comparison and memo-writing to ensure consistency and conceptual clarity.

3.4 Trustworthiness

Member checking was used to confirm the accuracy of participants' contributions. After the interviews were transcribed, participants reviewed their transcripts to verify that their perspectives were represented correctly [3]. No changes were requested. An audit trail was also maintained throughout the study, documenting key decisions during data collection and analysis [2]. This helped ensure transparency and supported the trustworthiness of the findings. To increase the trustworthiness of the findings, the analysis included detailed documentation of coding decisions and a review of coded data by an additional researcher to establish inter-coder reliability. Differences in interpretation were discussed until consensus was reached. These steps helped mitigate subjectivity and enhance the credibility of the findings, as recommended by Grbich [3]. Trustworthiness was additionally strengthened through methodological triangulation, as interview data were interpreted alongside course syllabi to corroborate faculty perspectives with instructional artifacts.

4 Findings: data story

The following study findings are presented as a narrative-based "data story." This approach weaves together the key themes that emerged from the interviews, incorporating representative quotes from each participant and capturing how faculty are actively adjusting the integration of AI into their teaching practices. Their experiences reveal a complex mix of inspiration, concerns, and ethical reflection. While the narrative is fictionalized in structure, it remains grounded in the actual interview data and was designed to present insights in a more accessible and cohesive format. The following data story highlights that AI is not simply a tool, but a force that is reshaping instructional practices in many ways.

The participants did not describe AI use as a clear linear journey. Their stories resembled navigating the halls of a busy science building, where each turn reflects a different influence like teaching values, institutional pressures, student needs, and the evolving rate of technological change.

Across interviews and syllabus review, six themes emerged: (1) student-centered teaching as a guiding value; (2) chemistry's emphasis on epistemic verification; (3) ethical negotiation of AI use; (4) AI as a supplemental instructional partner; (5) concerns about dependency and academic integrity; and (6) discipline-specific constraints on AI integration.

Student centered teaching stood out as a constant—the central hallway that anchored and guided every decision. The participants consistently emphasized that any use of AI must ultimately benefit students, not distract from learning, and not diminish the hands-on nature of chemistry education. Faculty expressed the need to remain flexible and stay critically aware of the power that AI holds, balancing innovation with skepticism. Ethical concerns identified by the participants included algorithmic bias, data privacy, and potential threats to academic integrity, which led the faculty to approach the use of AI with caution and deliberation.

Student retention was another common theme, and faculty were mindful of how AI might support or hinder student persistence. While some identified opportunities for personalized learning, others were skeptical of creating dependence or disengagement. When it came to AI as a teaching tool, opinions varied. While some welcomed the automation of repetitive tasks or enhanced feedback tools, others feared a loss of the human element in teaching, which they believed could prove detrimental to some learners. The participants also reflected on how AI fits specifically into chemistry instruction. The participants identified how they made unique decisions based on their course content, teaching style, and the needs of their students. Simulations, lab work, and data analysis were common entry points, but all participants noted that the integration of AI must be intentional and discipline appropriate.

The identified themes showed that faculty are not simply adopting or rejecting AI, they were continually negotiating its role in their work and considering how it could be a positive or negative tool in their teaching practices. The following data story captures this ongoing process and is shaped by intersecting goals of innovation, responsibility, and commitment to student learning. As Charmaz [17] suggested, narrative synthesis can bring clarity and depth to participant voices, helping researchers and readers better understand the lived experiences behind educational change. To support analytic transparency, all text enclosed in quotation marks reflects verbatim excerpts from participant interviews. Narrative descriptions linking quoted segments represent analytic synthesis derived from interview data and syllabus review. Interpretive statements are explicitly framed as analysis to distinguish them from participant voice.

4.1 AI in the AM—change is brewing

4.1.1 Links to theoretical framework and literature

This study drew on constructivist grounded theory methodology as articulated by Charmaz [17], which emphasizes meaning-making and theory development grounded in participants' lived experiences. An inductive analytic approach was employed to ensure that emerging themes reflected faculty members' authentic concerns, values, and instructional priorities rather than being shaped by pre-existing theoretical assumptions. Open coding was used to identify initial concepts such as AI as an instructional support, ethical uncertainty, and faculty sense-making amid rapidly evolving educational technologies [3]. These concepts emerged organically through sustained engagement with interview data and were refined through constant comparison as analytic relationships among themes became more visible [1, 2].

Importantly, the grounded theory approach adopted in this study was not intended to generate a formal or generalized theory, but rather to develop a context-sensitive conceptual explanation of how chemistry faculty negotiate AI integration within their

teaching practices. Chemistry is a distinctly hands-on discipline in which laboratory work, experimental verification, and error detection are central to both teaching and learning. Unlike fields in which learning outcomes may be demonstrated primarily through written or conceptual work, chemistry education routinely requires students to translate abstract concepts into physical experimentation. As a result, faculty decisions about AI use are shaped not only by pedagogical efficiency or innovation, but by concerns about whether AI-supported work preserves students' ability to interpret data, recognize experimental error, and demonstrate conceptual understanding in laboratory settings.

The themes identified in this study resonate with existing literature on AI integration in higher education, particularly within STEM disciplines, where scholars have highlighted both the transformative potential of AI and the ethical challenges it introduces [12, 18]. However, much of the existing literature treats STEM fields as analytically similar or focuses on AI use in non-laboratory contexts. By centering chemistry faculty perspectives, this study offers discipline-specific insight into how AI is evaluated through an epistemic lens that prioritizes hands-on learning, experimental accountability, and academic integrity. Although the participant group was small, the findings provide meaningful insight into how faculty in laboratory-based disciplines interpret and respond to AI technologies, contributing a nuanced, contextually grounded perspective to broader conversations about ethical and effective AI adoption in higher education.

Recent studies have documented positive learning affordances of generative AI in higher education, particularly in contexts emphasizing written reflection and peer feedback. Noroozi et al. [19], for instance, demonstrated that students using GenAI for peer feedback were able to generate more high-level suggestions, although many participants opted not to use AI due to concerns about reduced learning value. These findings align with broader calls for ethically guided and task-sensitive AI integration [11]. However, such work has largely focused on text-based learning environments. By contrast, the present study highlights how faculty in laboratory-intensive disciplines such as chemistry evaluate AI through a different epistemic lens—one centered on experimental reasoning, data validation, and hands-on competence.

4.1.2 Chemistry, conversations, and coffee meet at the intersection of tradition and innovation at a STEM university

The morning light filtered through the windows of Elijah's office, where stacks of chemistry textbooks mingled with digital screens displaying the latest research breakthroughs. His nameplate "Elijah, Ph.D. in Organic Chemistry" was still so new there was a shiny, almost mirror-like aspect to the gold facing. As he sat at his desk, sipping his coffee, his thoughts were focused on the day ahead. His office overlooked one of the many lawns scattered throughout the busy STEM university in the Southeastern United States. Today, as he prepared for class and reviewed ideas for a departmental meeting about integrating artificial intelligence into their curriculum, his thoughts turned to recent news about how the Nobel Prize had been awarded for a breakthrough in AI. The recognition of AI's potential in transforming industries seemed to validate his long-held belief that technology, when thoughtfully integrated, could revolutionize education in ways he was only beginning to understand. He had always insisted, "…what I try to do in my philosophy in general, is to make sure students understand the material verses memorizing

a bunch of old white guy names." That simple, powerful notion guided him: true learning meant building deep connections rather than simply absorbing facts.

Across the hall, Dr. Caldwell was walking hurriedly to his office, already 15 min late and juggling his keys and coffee tumbler. Unlike Elijah's calm and organized space, Caldwell's desk reflected the chaotic nature of a busy academic life. Papers, research proposals, his children's drawings of stick figures, and other miscellaneous items were strewn about, a testimony to his constant juggling of research and teaching responsibilities, not to mention family obligations. In one of his thoughtful moments, Caldwell had shared how much he admired the university. "I really liked this university's vision for balancing research and teaching. Being undergrad centric but with a strong and active graduate program available as well." For him, teaching was not about repeating content that students could easily find elsewhere. He often explained, "Teaching content knowledge is not, should not, be primarily what we do because there will always be other ways that they can get content… as the professor, I think it should be more oriented towards building understanding and connections with students." As he reviewed his notes, he envisioned a future where AI tools could handle routine queries, freeing him to focus on nurturing those critical conceptual connections and advancing both time management and sound pedagogy.

Further down the corridor, Professor Pax enjoyed the comfort of an office filled with well-loved books and vintage memorabilia. His space was a warm blend of tradition and modern insight, and his cherished "World's Best Chemist" coffee cup gifted to him by a student over a decade ago was always within reach. Pax had long seen himself as a mentor first and a teacher second. "I wanted to be somewhere in the middle where I could both mentor students directly and still teach and educate." His approach to teaching was both energetic and practical. He believed that technology could help him offer personalized support to each student. "I think we're going in a direction where it [AI] could take the engagement data you get from a student… and try and say, ohh, they're having trouble with questions from this section. I'm going to give them specialized support for this section." Pax's eyes sparkled at the possibility, imagining a classroom where technology empowered him to reach students more effectively.

Later that morning, the three professors convened for a departmental coffee break in the lounge. Over cups of caffeine and conversation, the discussion turned to the impact of technology on their teaching practices. Dr. Caldwell set the tone by observing, "It's easy to go back to what's comfortable and familiar in terms of course design. And most of us came through environments where we were lectured at and we took tests and we did homework and we made A's… that's what's familiar to us, that's what's comfortable to us." He paused thoughtfully before adding, "I think what was said earlier too, is that you think students are at such varying degrees. I think that really lends itself well to that answer because there's going to be some students more familiar [with AI technology] than others." His measured reflection encapsulated a sentiment shared by many in the room: the comfort of old practices often tempered the excitement of new possibilities. Elijah added, "It's kind of more about how do we best utilize it for the student… If they don't understand it and the AI gets something wrong, they need to know why it's wrong." While AI tools can assist in solving complex problems, the professors emphasized that students must understand the principles behind AI-generated solutions. Elijah went on to describe AI as a "useful and unignorable tool" and Caldwell agreed.

Professor Pax, always sensitive to shifts in educational paradigms, recalled some of the major changes he had witnessed over the years. "I think, it's going to inspire change in the same way any technology inspired change when they went from transparencies to PowerPoints. It changed when we had the pandemic, and everybody had to teach online. I noticed a sort of tectonic shift in how my students learn." His voice carried both the excitement and uncertainty of someone standing at the crossroads of old and new educational methods, and perhaps a bit of conflict. Elijah acknowledged that while integrating AI into education may be challenging, its utility is undeniable: "I think what we have to do is almost treat it like we treat calculators, right?"

The conversation soon turned to ethics, a topic that weighed heavily on the minds of all three. With a calm yet serious tone, Elijah stated, "I don't see anything wrong with using AI if it's done in a way where it's a supplemental tool." His words, simple but steadfast, resonated with the group as they considered the delicate balance between innovation and responsibility. Caldwell was quick to raise concerns about the origins of AI content. "The large language models, they were trained on a tremendous number of other people's work. Not citing that properly or representing the AI's output accurately, well, I think that's the problem." He stressed the need for accountability in the use of such tools. "So tell me how much of this five page lab procedure that you are turning in? Tell me what the AI contributed. Include your prompts to the AI and the raw output. Now show me the finished version." Caldwell's practical approach underscored the common concern that the benefits of AI must not come at the expense of academic integrity and proper citation should be included when used.

Pax, recalling his own initial reservations, admitted, "My first impulse was, oh, it's going to be an easy way for them [students] to cheat." He paused, allowing the weight of his words to settle before continuing, "You have no way of knowing if they're using it… it's mostly a philosophy statement, which means it's not really a policy at all," referring to his *no use allowed* AI syllabus statement; one of three choices encouraged by the university (i.e., full use allowed, limited use allowed, and no use allowed). In this exchange, the participants acknowledged the ethical tightrope they have to walk in educational settings, protecting academic integrity while embracing technological advances in the rapidly evolving educational landscape.

Caldwell grew animated as he shared his experimentation with AI. "There are tools that, if you give them a reasonably well-detailed prompt, they can actually generate lecture notes on a topic. And I've looked through them, it's actually an improvement from my own set of guided notes. This summer I tried an assignment where I used AI to generate a fresh set of case studies to teach from." His voice carried both excitement and cautious optimism as he envisioned the efficiency and creativity such tools could unlock. He added, "And I'm hoping to have time this semester to have them [students] partner up with an AI to work toward even more lab implementations."

Pax, always on the lookout for ways to personalize his teaching, contributed enthusiastically: "If I fed in my entire question library to AI and said I want you to assign the topics that are relevant for each of these questions and try and tie these together. If they could do that, then that would be a huge time saver." His words painted a picture of a future classroom where routine tasks were streamlined, allowing him more time to engage directly with students and tailor his support to their specific needs. However, he went on to add, "I have a difficult time interacting with AI like it's a person. And it might

just be how I interpret human interactions versus AI interactions, I don't know. But I also don't chat too much via text in short bursts like that. So it might be that it's more human than those, but it doesn't line up for me I guess," he stated as his voice trailed off at the end and he shook his head emphatically. Elijah interjected by stating, "Some professors are trying to prohibit the use of AI, but I think that's kind of silly because it's just going to become a thing that everyone's going to use. I think that's just the inevitable sort of path of it. Because it's a useful tool and so people will keep using it." Pax and Caldwell raised their eyebrows and looked at each other in agreement.

Later, the conversation shifted to the broader implications of AI for student success and program appeal. Elijah reminded his colleagues that the integration of AI was not merely about keeping up with technological trends but how these innovations might further enhance student success and the overall appeal of their academic programs. Elijah pointed out that embracing AI wasn't just about modernizing the curriculum, it was also about preparing students for a future where digital literacy and technological competence would be crucial. "We have to figure out how they will be expected to use AI tools when they graduate, when they're on the job even 5 or 10 years down the line," he asserted firmly. For Elijah, the ultimate measure of progress was how well students were prepared for the real world. Pax agreed with a thoughtful nod. Elijah continued, "And if there are tools that I can provide that help drive us towards better communication, then they're going to intrinsically make the barrier to education lower. It's kind of where I'm at." He went on, "If they could get instantaneous and reliable support. That would be excellent. That would lower the barrier to them getting full meaningful feedback." The vision was clear: by harnessing AI, they could improve not only classroom learning but also the overall student experience. Elijah agreed, "We have to use it in tandem with our own understanding of material."

As the coffee break drew to a close, the professors reflected on the current state of AI in their field. Caldwell offered a reflective observation, "I would say that we are just kind of poking at it, most of us have not really given it a try for very much except maybe as a curiosity." He added with a hint of both uncertainty and optimism, "But it's getting to a point where it's useful, but we don't really know how to use it or how to teach with it. Or how to teach them to use it." Pax, ever thoughtful, looked out the window and pondered aloud, "So when I look at it in our view and I say, well, how can I adapt what I'm doing to this emergent tool, and as I know what its new capabilities are continue to change because I can't really even spot where it's going in terms of the tools that are available of it." Then, with characteristic humor and depth, he added, "And oh, gosh, sorry, we're going to be answering this with philosophy questions. Every human problem is a new problem, and the tools we have to answer those problems aren't necessarily new solutions. So, when I see in politics something like, oh, we've got to find a way to feed the hungry people. We have tons of ways to feed hungry people. People might have had a different reason for being hungry. So, the problem is new when it's a human problem, always. We might get new solutions from this instead of just thinking about new problems. I think it's worth thinking about how this might be something that's new in the toolkit that we haven't had before."

In that quiet moment after their discussion, the three professors recognized that the future of chemistry education lay at the intersection of tradition and innovation. Their conversation had not provided all the answers, but it had ignited a shared vision. They

knew that while the promise of AI was vast, its true value would only emerge when it was harnessed thoughtfully and responsibly. The day passed by for the three professors. Elijah taught his class with his usual emphasis on deep understanding, Caldwell, despite his late arrival, delivered his lesson with a focus on helping students make connections, and Pax embraced his role as both a mentor and an innovator in his field as he drank from his same well loved "Worlds Best Chemist" chipped coffee cup gifted to him by that student over a decade ago.

As the professors prepared to leave campus, each carried a personal sense of the challenges and opportunities that lay ahead. Elijah left his office with renewed determination to integrate AI into his teaching practices in ways that would deepen student understanding without sacrificing the human connection that was essential to learning. Dr. Caldwell, amid the organized chaos of his desk, resolved to explore AI further while ensuring that its use would be held to rigorous ethical standards. Professor Pax walked away with both caution optimism and a commitment to preserving the time-honored tradition of mentoring. As faculty members departed campus, their individual reflections underscored the deeper implications of this study. While their educational approaches and institutional contexts varied, each participant grappled with the opportunities and responsibilities that artificial intelligence (AI) introduced into their teaching practices, as well as their understanding of the complex, evolving nature of faculty engagement with AI.

5 Discussion

Recent empirical work has demonstrated that generative AI can support student learning in structured, text-based tasks such as peer feedback provision. For example, Noroozi et al. [19] found that graduate students using GenAI during peer feedback tasks were able to provide more high-level suggestions, although many students deliberately chose not to use AI in order to preserve their own learning. While these findings highlight the pedagogical value of GenAI in feedback-oriented contexts, they also underscore the importance of task structure and epistemic goals. In contrast, chemistry education is deeply rooted in experimental practice, data interpretation, and error detection, raising distinct concerns about how AI-generated input may influence students' development of disciplinary judgment.

This study contributes a discipline-specific understanding of faculty engagement with AI by showing that chemistry instructors evaluate AI less in terms of efficiency gains and more in terms of epistemic risk. Unlike findings from broader faculty studies, chemistry faculty in this study framed AI adoption as a negotiated human–AI collaboration shaped by experimental accountability, error sensitivity, and pedagogical responsibility.

Rather than reiterating familiar tensions such as innovation versus integrity, this analysis reveals how those tensions are actively mediated through chemistry's disciplinary identity. Faculty decisions about AI use were informed by concerns about whether AI-supported work preserved students' ability to evaluate correctness, interpret experimental outcomes, and justify conclusions.

5.1 Scope and transferability

Given the small, homogeneous sample, the findings are not intended to be statistically generalized. Instead, this study offers analytically transferable insight into how faculty

operating within experimentally grounded disciplines may approach AI integration under similar institutional and epistemic conditions.

6 Conclusion

In response to the research questions, the data story clearly highlighted the central role ethics played in how faculty perceived AI in education. Throughout the narrative, it became apparent that chemistry professors were not merely concerned with *what* AI could do, but with what it *should* do in the context of a chemistry classroom. This was particularly evident when discussing the ethical implications of AI's use, such as ensuring student understanding and accessibility.

The first research question asked: *How do chemistry faculty view the role of AI in student learning within their discipline?* As a whole, the faculty did not see AI as a shortcut to understanding, but rather as a supplemental tool that could help students engage with material when used thoughtfully and ethically. This came through especially in Pax's conversation, where he framed AI as a powerful new resource but also warned that it could never replace the hands-on process required to gain deeper understanding into chemistry. Elijah also agreed that AI was a useful tool, and students would probably be expected to use it in the future. Caldwell shared that chemistry faculty were "just kind of poking at it, most of us have not really given it a try for very much except maybe as a curiosity, and most of us probably don't understand all the ways that it can be used for teaching," and that although he initially believed that he would "never want students to use AI" in his class, he is starting to "see some value in it."

The second research question asked: *What are the perceived benefits and challenges of integrating AI technologies into chemistry instruction?* The data showed a broad spectrum of views, with some faculty being optimistic and others more skeptical. However, all three recognized that challenges around maintaining academic integrity and fostering real understanding would not disappear with the implementation of AI. Caldwell's reflections highlighted these tensions, explaining that he saw benefits in using AI for tasks like drafting lab reports or offering practice problems, but he also voiced strong concerns about overreliance and the erosion of critical thinking skills. Pax also had concerns about potential negative impacts on student learning, stating that "for them [students], the grade is the main thing, but for me, the learning is the main thing." Pax also shared that "the place I think it [AI] will impact the most is lab reports. I think they'll have AI write a lab report for them first." Elijah also shared concerns about the integration of AI in chemistry classrooms, but was also hopeful, sharing that "it's kind of more about how we best utilize it [AI] for the student, because they still need to understand it some way, even if it can solve it for them. If they don't understand it, and the AI gets something wrong, they need to know why it's wrong." The faculty also recognized the benefits of AI and the need to incorporate it into their lessons, but acknowledged they were still learning the best way to accomplish its integration. Elijah shared that "I don't think we've seen the AI systems evolve enough to really understand their use." Elijah also explained that "we can ignore it, but in reality, everyone else is going to be using it. And so, we sort of have to monitor how AI is progressing and take the most advantage of it as we can. Because otherwise we're just going to be doing our students a disservice where, you know, they come in and four years from now AI has evolved, but we haven't been able to use it. And now, all of a sudden, all the companies are like, yeah, we just use this.

It's easy. So, then you have to figure out the best way to incorporate these future AI tools into our classrooms." Caldwell shared a similar concern regarding the preparation of students for the workforces, sharing "I think figuring out what how will they be expected to use AI tools when they graduate and get a job or maybe even 5 or 10 years after that and nobody knows what AI tools are going to look like 5 or 10 years from now. And so, it's hard for us to offer that training when we don't really know ourselves what it can do, what it will become able to do."

The third research question asked: *How do faculty navigate the ethical considerations associated with AI tools in their teaching practices?* Issues of academic integrity, fairness, and responsible technology use were tightly woven into the way professors approached AI integration. The faculty did not just worry about students misusing AI, they also thought deeply about how their own teaching practices needed to adapt in a way that stayed true to their values. Pax pointed to the root of this issue when he noted that ethics were not a separate conversation; they were built into every decision faculty made about when, how, and whether to use AI in their courses. Eliah and Caldwell also both made statements that students should be citing AI within their work to adhere to ethical standards.

Another key finding that emerged from the data was the desire of faculty for more personalized professional development. While this finding was not explicitly illustrated in the fictionalized data story, it surfaced organically in the interview conversations, revealing a shared faculty interest in learning opportunities that felt directly relevant to their discipline. Rather than attending broad, generalized training, participants expressed a preference for sessions led by peers within their own field. They imagined conference presentations or hands-on workshops facilitated by chemistry educators who had already experimented with AI tools in their own classrooms. They believed that these types of sessions would carry more weight because they would be grounded in real instructional challenges, classroom examples, and shared language. Elijah discussed the benefit that an outside expert might provide regarding professional development, when he shared "we need someone who is an expert in these things to really tell us what is the best way to use these tools to best benefit the students and not just be like, oh, you can't use these. Because that's just going to be again, a disservice to the students." Elijah also shared that this expert should be "someone who understands the needs of the students [and] understands how these tools will get utilized beyond the university level. Because that's where they're going to end up eventually, and we need to make sure they're prepared for that and that we're prepared to help them in that process."

At the same time, participants acknowledged that good pedagogy often extended beyond discipline-specific boundaries. One faculty member shared that he had recently joined a cross-disciplinary book club focused on teaching with AI, which helped him reflect on instructional strategies from a broader perspective. Another mentioned plans to apply for an internal grant centered on AI use in the classroom, signaling proactive interest in furthering his practice. These examples suggest there remains an openness to collaboration when the content aligns with immediate instructional goals. As such, future professional development opportunities, whether delivered by discipline-specific peers or by campus instructional designers, could benefit from being tailored, timely, and grounded in both pedagogical best practices and real-world teaching contexts.

The findings of this study add to the expanding conversation in literature that faculty should not simply be expected to adopt new technologies without context or training. Faculty reflections demonstrated that AI use often sparked deeper pedagogical conversations about purpose, alignment, and what learning should look like in a technologically evolving environment. Their decisions were rarely made in isolation but were informed by discussions with their peers, departmental culture, personal teaching philosophies, and the perceived needs of their students.

Future research should continue to explore how disciplinary differences shape faculty experiences with AI, especially as these tools become more embedded over time. Longitudinal studies tracking shifts in faculty perspectives would offer valuable insight into how trust, innovation, and caution evolve together. It would also be beneficial to incorporate student perspectives, building a more complete understanding of how AI shapes teaching and learning. The study findings revealed that the integration of AI in education is not simply a matter of adoption, but an ongoing, multifaceted discussion—where innovation and discipline-specific identity are deeply intertwined. Faculty in this study demonstrated that their engagement with AI was not only shaped practical considerations but also philosophical reflections on the future of education in their field.

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M.M. developed the conceptual design, conducted and transcribed participant interviews, analyzed the data, and drafted the initial manuscript. D.P. performed secondary data analysis and contributed revisions and additions. All authors collaboratively reviewed multiple drafts and approved the final version.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Institutional Review Board of Tennessee Tech University (6/3/2024/No. 2808). Informed consent was obtained from all individual participants included in the study.

Consent for publication
Not applicable. This study does not include any individual person's data or images that could lead to the identification of a study participant.

Competing interests
The authors declare no competing interests.

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